

Aviral Goel

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Professional Summary

B.Tech (AI & Data Science, 2027) undergraduate with a 9.1 CGPA, skilled in Python, deep learning, computer vision, NLP, and multi-agent LLM systems. Experienced in building and deploying intelligent systems across speech synthesis, anomaly detection, and financial analytics, with production ML pipelines and agentic architectures (LangGraph). Brings speech research experience from Jio Platforms and GPU/HPC infrastructure work at Yotta Data Services on NVIDIA-powered clusters.

Education

KJ Somaiya College of Engineering, Mumbai <i>B.Tech in Artificial Intelligence & Data Science; CGPA: 9.1/10, SGPA: 9.94 (Sem VI)</i>	Jul 2023 – May 2027
CBSE (12th) – NBCS (80.4%)	Jun 2021 – May 2023
CBSE (10th) – DPS School (88.8%)	Jun 2015 – May 2021

Experience

Jio Platforms Limited <i>Speech & TTS Research Intern</i> - Benchmarked six TTS architectures for zero-shot voice cloning with a research team across 11 Indic languages (OmniVoice). - Eliminated five engineering blockers and audited 300+ flagged issues in a Hindi TTS training dataset.	May 2026 – Jun 2026
Yotta Data Services <i>Operations Intern</i> - Provisioning and monitoring NVIDIA H100/Blackwell GPU clusters on Yotta’s Shakti Cloud alongside operations engineers. - Implementing load-balancing and node-health monitoring across GPU clusters to maintain uptime.	Jul 2026 – Aug 2026
Firetech Industries (Freelance) <i>Library Management System (Leela’s Library)</i> - Engineering a Library Management System (Python, SQL) automating cataloging, checkout, and OTP-based authentication. - Digitizing a fully paper-based catalog of 960+ books into a searchable, automated system.	Jul 2026 – Sep 2026
KJSSE (In-house Internship, KJ Somaiya) <i>AI-Powered Anomaly Detection in EV Charging Infrastructure</i> - Designed an anomaly detection framework (Isolation Forest, LSTM, Autoencoder) across 1,000+ EV transactions at 94% accuracy. - Engineered a scalable pipeline processing 10,000+ points/hour, cutting false positives by 35%.	Aug 2025 – Oct 2025

Projects

ResiliNet – Financial Systemic Risk Simulator <i>Python, GNN, FastAPI (Hackathon)</i> - Built the ML/backend core of a graph-based risk simulator using a GAT to model interbank cascade failures across 50 nodes. - Implemented the backend REST API with a teammate, serving real-time risk scores at 23% higher accuracy over baseline.	GitHub Live	Mar 2026
MultiCodeReviewAgent <i>LangGraph, FastAPI, Docker, Ollama, Groq API, React</i> - Architected a multi-agent code review system with four LLM agents (Security, Style, Logic, Judge) via LangGraph fan-out. - Developed a dual-inference pipeline (Ollama/Llama 3.2 + Groq fallback), validated against a 100+ line test codebase.	GitHub Live	Jul 2026
StockSage <i>React, FastAPI, Groq (LLaMA 3.3 70B), FinBERT, SQLite</i> - Developed a full-stack AI stock platform with 30 FastAPI endpoints spanning live data, portfolio tracking, and alerts. - Integrated an LLM chat assistant (Groq LLaMA 3.3 70B) and a FinBERT sentiment engine feeding a Buy/Hold/Sell recommender.	GitHub Live	Feb 2026
Driver Drowsiness Detection System <i>Python, OpenCV, Mediapipe, FastAPI</i> - Designed a real-time CV system using Mediapipe’s Face Mesh and calibrated EAR thresholds at 92% accuracy and 95% precision. - Built a threaded video pipeline sustaining 30 FPS at 40ms latency, reducing alert trigger time to 0.8 seconds.	GitHub Live	May 2026

Technical Skills

Languages & ML: Python, SQL, DSA (LeetCode), TensorFlow, PyTorch, Scikit-learn, Deep Learning, NLP, LangGraph, GNNs
Data Science & Visualization: Pandas, NumPy, Tableau, Power BI, Excel, EDA, Statistical Analysis, Data Engineering
Backend, DB & Tools: FastAPI, REST APIs, Docker Compose, GPU/HPC Ops, React, MySQL, PostgreSQL, MongoDB, Git
Core Competencies: Collaboration, Communication, Problem-Solving, Leadership, Research & Development

Certifications & Achievements

IBM Data Science Professional Certificate – IBM | Database Structures and Management with MySQL – Meta (Coursera)
Shortlisted – SwarmGuard, Blockchain & Agentic AI Hackathon | ResiliNet – KJSCE Datathon | SIH 2023 & 2024 | 5+ Hackathons